INCH-POUND

MIL-A-53089A
AMENDMENT 1
1 June 1992

MILITARY SPECIFICATION

AIR CONDITIONERS: VERTICAL AND HORIZONTAL, COMPACT, MULTIPLE POWER INPUT

This amendment forms a part of MIL-A-53089A, dated 17 January 1992, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, line 10, under type II, add:

"A - 9,000 Btu/hr - 208/230 V, single/3-phase, 50/60/400 Hz."

PAGE 2

2.1.1, under SPECIFICATIONS, FEDERAL, delete "PPP-C-843" in its entirety and add:

"A-A-1898 - Cushioning Material, Packaging, (Cellulosic, Water Absorbent)."

Under SPECIFICATIONS, MILITARY, add:

"MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant."

PAGE 3

2.1.2, line 19, add:

"TA13229E9900 - Air Conditioner, Horizontal, Compact, 9,000 Btu/hr, 208/230 Volt-AC, Single/3-Phase, 50/60/400 Hertz."

PAGE 6

3.1, line 8, under "Type II - Horizontal", add:

"Size A - 9,000 Btu/hr - TA13229E9900"

AMSC N/A FSC 4120

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3.5.2, table I, delete in its entirety and substitute:

"TABLE I. Cooling capacities.

Unit	Condition	Ambient temperature (dry bulb)	Evaporator inlet temperature		Minimum net sensible cooling capacity	Minimum net total cooling capacity
			Dry bulb	Wet bulb		
		°F	°F	°F	Btu/hr	Btu/hr
Type I						
Size B	A	120	90	75	5,600	9,350
	B	95	80	67	6,000	9,350
Size C	A	120	90	75	12,100	18,500
	B	95	80	67	14,000	21,500
Size D	A	120	90	75	22,680	37,800
	B	95	80	67	24,570	37,800
Type II						
Size A	A	120	90	75	8,000	10,000
	B	95	80	67	8,000	10,500
Size B	A	120	90	75	14,000	18,500
	B	95	80	67	14,700	19,500
Size C	A	120	90	75	30,000	41,000
	B	95	80	67	32,000	40,000

Note: Data shown for 0-inch water gage (WG) (free air) external static pressure on the evaporator for type I units and 0.25 inch WG for type II units."

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3.5.3, table II, delete in its entirety and substitute:

TABLE II. Condensing and suction pressures (psig).

Unit	Operating conditions (see 3.5.1.1)			Operating conditions (see 3.5.2)	
	Condensing pressure (max)	(min)	Suction pressure (min)	Condensing pressure (max)	Suction pressure (min)
Type I					
Size B	470	-	50	405	65
Size C	470	-	50	440	65
Size D	470	105	50	430	65
Type II					
Size A	470	160	50	425	50
Size B	470	160	50	430	50
Size C	470	160	50	430	50

3.5.6, table III, delete in its entirety and substitute:

"TABLE III. Airflow, heating capacity and power consumption.

Unit	Evaporator Airflow (minimum SCFM) (1)	Heating (minimum Btu/hr) (2)	Power Consumption (maximum KW) (3)	Power Factor (minimum) (4)
Type I				
Size B	335	6,000	3.6	0.9
Size C	570	12,000	6.5	0.9
Size D	1,250	28,600	11.0	0.83
Type II				
Size A	370	7,000	4.2	0.71
Size B	580	13,400	6.5	0.9
Size C	1,380	31,200	13.5	0.9

3.5.7, line 9, after "Type II", add:

"Size A - 208/230 V, single/3-phase, 50/60/400 Hz."

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3.7, delete in its entirety and substitute:

"3.7 Noise. The steady-state noise levels produced by the air conditioners shall conform with the requirements of MIL-STD-1474, except as indicated herein when tested in accordance with 4.5.2.26.

a. The steady-state noise level on the evaporator side of the air conditioners shall not exceed MIL-STD-1474, table II, category F [65 dB(A)] at a one meter hemispherical perimeter from the center of the evaporator surface. Alternate preferred speech interference levels (PSIL) shall not apply.
b. The steady-state noise level of the condenser side of the air conditioners shall not exceed MIL-STD-1474, table II, category D [85 dB(A)] at a one meter hemispherical perimeter from the center of the condenser surface of the air conditioner. Octave band limits of MIL-STD-1474 shall not apply.
c. Where the noise limits specified above can be documented as being clearly beyond the state-of-the-art, selection of another noise limit shall be made by the procuring activity.
d. The provisions MIL-STD-1474, 4.2 and 4.3 shall be provided if and only if procedures stated in MIL-STD-1474, 5.1.1.3.2 of have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limits is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.2 and 5.1.2.1.4."

PAGE 12

3.10.4, table IV, delete in its entirety and substitute:

"TABLE IV. High and low cutout pressures and fan speed cut-in pressure.

Unit	Compressor high-pressure cutout (±10 psig)	Compressor low-pressure cutout (±5 psig)	Fan high-speed cut-in (psig)	Fan low-speed cut-in (psig)
Type I				
Size B	480	25	391 ±31	285 ±17
Size C	480	7	-	-
Size D	480	7	-	-
Type II				
Size A	480	15	-	-
Size B	480	7	-	-
Size C	480	15	-	-

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PAGE 16

3.16, table VI, delete in its entirety and substitute:

"TABLE VI. Weight.

Type	Size	Maximum Weight (lb)
I	B	200
	C	280
	D	460
II	A	200
	B	290
	C	435

3.17, line 4, after "MIL-C-46168", add "or MIL-C-53039".

PAGE 18

4.3.2, line 2, delete "table VIII" and substitute "table VII".

PAGE 33

4.5.2.19.4, table XII, delete in its entirety and substitute:

"TABLE XII. Minimum temperature drop and maximum current draw. 1/

Unit	Input Power	Temperature difference °F (minimum)	Current draw Amps (max)
Type I			
B	208/230V, single/3-phase, 60 Hz	17	15
C	208V, 3-phase, 60 Hz	20	24
D	208V, 3-phase, 60 Hz	18	32
Type II			
A	108/230V, single/3-phase, 60 Hz	18	15
B	108/230V, single/3-phase, 60 Hz	20	37 2/
C	208V, 3-phase, 60 Hz	20	45

1/ For all 3-phase air conditioners, amperage reading shall be the average of the three phases.
2/ For all 3-phase operation, current drawn shall not exceed 25 amps."

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PAGE 36

4.5.2.26, line 2, after "MIL-STD-1474", insert ", 5.1.2".

PAGE 38

5.2.1.1.a, lines 5 and 6, delete "PPP-C-843" and substitute "A-A-1898, grade II, class B, style and size optional".

5.2.1.1.b, line 5, delete "PPP-C-843" and substitute "A-A-1898, grade II, class B, style and size optional".

PAGE 39

5.3.2.1, line 6, delete "size 3/4 X 0.015 or equivalent" and substitute "coated-finish, size 3/4 X 0.015 and ASTM D 4675".

PAGE 42

6.8, under "Type II - Horizontal, add:

"A 1 - 230V, single phase, 50/60 Hz	A - 230V, single phase, 50/60 Hz
2 - 208V, 3-phase, 50/60 Hz	- 208V, 3-phase, 50/60 Hz
3 - 208V, 3-phase, 50/60 Hz"	

PAGE 49

Figure 6, note number 4, delete in its entirety and substitute:

"4. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, coated-finish, size 3/4 X 0.015 and ASTM D 4675."

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DLA - GS

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